BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 23-G-0225 & 23-G-0226

September 1, 2023

Prepared Testimony of:
Staff Forecasting
and Inflation Panel

Sean Isakower
Principle Economist

Daniel Gadomski
Utility Analyst II

James Creaser
Economist Trainee II

Office of Regulatory Economics
State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Forecasting and Inflation Panel, or the Panel, please state your names, employer, and business address?

A. Our names are Sean Isakower, Daniel Gadomski, and James Creaser. We are employed by the New York State Department of Public Service, or the Department. Our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Dr. Isakower, what is your position in the Department?

A. I am employed as a Principal Economist in the Office of Regulatory Economics.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Science in Economics from The College of New Jersey in 2008, and a Ph.D. in Economics with a concentration in Industrial Organization in 2014 from Northeastern University. Prior to joining the Department in 2014, I taught statistics at Northeastern University.

Q. Please briefly describe your current responsibilities at the Department.

A. My main responsibilities as Principal Economist

are providing economic and analytical expertise on energy and telecommunications issues, including the current data initiatives, storage initiatives, and sales forecasting for the purpose of rate cases.

Q. Have you previously testified before the New York State Public Service Commission, or the Commission?

A. Yes, I have testified in numerous rate proceedings. I testified in Case 14-E-0318 regarding Central Hudson Gas & Electric Corporation, or Central Hudson; 15-E-0283, 15-E-0285, 19-E-0378, and 19-E-0380 regarding New York State Electric & Gas Corporation, or NYSEG, and Rochester Gas & Electric Corporation, or RG&E; Cases 17-E-0283 and 20-E-0380 regarding Niagara Mohawk Power Corporation d/b/a National Grid, or Niagara Mohawk; and most recently I testified in NYSEG/RG&E's most recent rate proceeding, Cases 22-E-0317, 22-G-0318, 22-E-0319, and 22-G-0320.

Q. Was your testimony in the previously referenced cases all related to sales forecasting?

A. Although most are, the latest two cases were

not. For Niagara Mohawk's 20-E-0380 rate case I testified to energy storage projects on the Staff Clean Energy Programs Panel, and in NYSEG/RG&E's latest rate cases 22-E-0317 et al., I testified on a number of topics including the Integrated Energy Data Resource and NYSEG/RG&E's Flexible Interconnection Solution, or FICS.

Q. Mr. Gadomski, what is your position in the Department?

A. I am a Utility Analyst 2 in the Office of Regulatory Economics.

Q. Please briefly state your educational background and professional experience.

A. I received a Bachelor of Arts Degree in Economics from the State University of New York at Albany in 2014. I have been employed by the Department since 2014.

Q. Please briefly describe your current responsibilities at the Department.

A. My responsibilities at the Department include analyzing inflation and sales forecasting issues in rate proceedings.

Q. Have you previously testified in proceedings before the Commission?

A. Yes. I provided testimony on compensation and benefits issues in Cases 16-G-0059 and 19-G-0309 regarding KeySpan Gas East Corporation d/b/a National Grid, or KEDLI; Cases 16-G-0058 and 19-G-0310 regarding The Brooklyn Union Gas Company d/b/a National Grid NY, or KEDNY; Cases 16-W-0130, 19-W-0168, and 19-W-0269 regarding Suez Water New York, Inc., SUEZ Water Westchester Inc., and SUEZ Water Owego Nichols Inc., or SUEZ; Case 16-G-0257 regarding National Fuel Gas Distribution Corp., or National Fuel; Case 16-W-0259 regarding New York American Water Co., or NYAW; Case 16-G-0369 regarding Corning Natural Gas Corporation, or Corning Natural Gas; Cases 17-E-0238 and 17-G-0239 regarding Niagara Mohawk; Cases 17-E-0459 and 17-G-0460 regarding Central Hudson; Cases 18-E-0067, 18-G-0068, 21-G-0073, and 21-E-0074 regarding Orange and Rockland Utilities, Inc., or O&R; and Cases 19-E-0065 and 19-G-0066 regarding Consolidated Edison Company of New York, Inc., or Con Ed. I also provided testimony on both compensation and benefits issues and gas sales forecasting in Cases 20-G-0101 and 21-G-0394 regarding Corning

Natural Gas; and Cases 20-E-0428 and 20-G-0429 regarding Central Hudson. I have provided testimony on inflation and sales forecasting issues in Cases 22-E-0317, 22-G-0318, 22-E-0319 and 22-G-0320 regarding NYSEG/RG&E; Case 21-G-0577 regarding Liberty Utilities, or St. Lawrence Gas Corporation; Cases 20-G-0101 and 21-G-0394 regarding Corning Natural Gas; and Cases 20-E-0428 and 20-G-0429 regarding Central Hudson

Q. Mr. Creaser, what is your position in the Department?

A. I am employed as an Economist Trainee II in the Office of Regulatory Economics.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Science in Business Economics and Communications from The State University of New York at Oneonta in May 2020. From May 2020 through March 2022, I worked at Homestead Funding Corp. as a loan officer. I have been employed by the Department since April 2022.

Q. Please briefly describe your current

responsibilities at the Department.

A. My responsibilities include assisting in sales forecasting in rate proceedings and maintaining a Herfindahl-Hirschman Index for the wholesale generation market.

Q. Have you previously testified before the Commission?

A. Yes. I have testified before the Commission in Rate Cases 22-E-0317 and 22-G-0318 regarding NYSEG; Cases 22-E-0319 and 22-G-0320 regarding RG&E; and Case 22-S-0659 regarding Con Ed's steam service. I testified on the topic of sales forecasting for each of these cases.

Q. What is the purpose of the Panel's testimony in these proceedings?

A. In our testimony, we will describe the meter count and natural gas sales forecasts submitted by the Gas Load Forecast Panel for KEDNY and KEDLI, collectively referred to as the Companies. We will then discuss our recommendations regarding the meter count and natural gas sales forecast for the Rate Year, defined as Fiscal Year, or FY, 2025, or the 12 months ending March 2025. The goal of our

forecasting methodology is to minimize forecasting errors and reduce rate instability. We will also address the Companies' calculation of a general inflation rate that is used to escalate cost of service elements except for labor, or non-labor inflation calculation. This calculation was presented in Exhibit____(RRP-8) of the Companies' Revenue Requirements Panel, or RRP.

Q. In your testimony, will the Panel refer to, or otherwise rely upon, any information produced during the discovery phase of these proceedings?

A. Yes. We will refer to, and have relied upon, several responses to Staff's Information Requests, or IRs. These responses are included in Exhibit___(SFIP-1). We will refer to them by the designation given to them by Staff, for example, DPS-100.

Q. Is the Panel sponsoring any other exhibits?

A. Yes. We are also sponsoring five additional exhibits, for a total of six exhibits.

Q. Please briefly describe these additional exhibits.

A. Exhibit___(SFIP-2) and Exhibit___(SFIP-3) are

summaries of this Panel's forecasts for delivery volumes and meter counts along with comparisons with the Companies' forecasts for KEDNY and KEDLI, respectively. Exhibit___(SFIP-4) and Exhibit___(SFIP-5) show the coefficient estimates and summary statistics associated with our forecasting models of KEDNY's and KEDLI's meter count and sales, respectively. Exhibit___(SFIP-6) contains this Panel's calculations of inflation for the 12 months ending December 31, 2022, referred to as the Historic Test Year, or HTY; the 12 months ending March 31, 2025, referred to as the Rate Year or FY 2025; the 12 months ending March 31, 2026, which the Companies referred to as Data Year 1, or DY1, or what we refer to as FY 2026; the 12 months ending March 31, 2027, which the Companies referred to as Data Year 2, or DY2, or what we refer to as FY 2027; and the 12 months ending March 31, 2028, which the Companies referred to as Data Year 3, or DY3, or what we refer to as FY 2028, based on the latest available Blue Chip Economic Indicators forecasts of Gross Domestic Product - Price

Index inflation published on August 11, 2023.

Companies' Forecast Summary

Q. What are the Companies' Rate Year gas delivery volume forecasts?

A. On page 1 of Exhibit___(GLF-4A) and Exhibit___(GLF-4B) of the Gas Load Forecast Panel's direct testimony, the Companies forecast a gas delivery volume of 1,689,204,366 therms for KEDNY and 1,374,228,372 therms for KEDLI, respectively.

Q. What are the Companies' Rate Year meter count forecasts.

A. On page 1 of Exhibit___(GLF-5A) and Exhibit___(GLF-5B), the Companies forecast meter counts of 1,285,927 for KEDNY and 642,298 for KEDLI, respectively.

Q. Did the Companies include any adjustments to account for demand side management programs in the Companies' service areas and the impact of local laws intended to reduce gas usage?

A. Yes. The Companies included adjustments related to New York City Local Law 97 which sets greenhouse gas, or GHG, emission limits on buildings in the City of New York over 25,000

square feet beginning in 2024; and Local Law 154, or LL 154, which prohibits installation of gas systems or equipment in newly constructed buildings in the City of New York less than seven stories tall beginning in 2024, and prohibits such gas use in newly-constructed buildings greater than seven stories beginning in 2027 with exceptions for certain building types. The Companies also included adjustments to reflect energy efficiency and electrification savings.

Q. Did the Companies include an adjustment related to the All-Electric Building Act, or the AEB Act?

A. No. The AEB Act was adopted as part of the FY 2024 New York State Budget. The Companies did not consider the AEB Act since the Companies filed their Rate Cases prior to the passage of the FY 2024 New York State Budget.

Q. Please briefly summarize the requirements of the AEB Act.

A. The AEB Act prohibits the installation of gas systems or equipment in new construction buildings up to seven stories, with exceptions

for commercial or industrial buildings greater than 100,000 square feet, beginning in 2026, and in all new buildings from 2029 onwards. The AEB Act does not restrict or prohibit oil-to-gas conversions in existing buildings or low-use upgrades to heating.

Q. Did the Companies provide an update to their Rate Year meter count and gas delivery forecasts in their Corrections and Updates filing on June 30, 2023?

A. Yes.

Q. Please describe the Companies' Updates and Corrections to their Rate Year gas delivery volume forecasts.

A. On page 1 of Exhibit___(GLF-4ACU) and Exhibit (GLF-4BCU), the Companies forecast a gas delivery volume of 1.665 billion therms for KEDNY and 1.3 billion therms for KEDLI, respectively, in FY 2025.

Q. Please describe the Companies' Updates and Corrections to their Rate Year meter count forecasts.

A. On page 1 of Exhibit___(GLF-5ACU) and Exhibit___(GLF-5BCU), the Companies forecast

meter counts of 1,287,896 for KEDNY and 637,032 for KEDLI, respectively, in FY 2025.

Q. Did the Companies' updated forecasts account for the AEB Act?

A. Not directly, but the Companies' initial and updated forecasts do consider LL 154. On page 12 of the Companies' Gas Load Forecast Panel's Corrections and Updates testimony, the Companies explained: "Given that LL 154 preceded the AEB Act, and that the two pieces of legislations are not drastically different in terms of timelines and exemptions, the Companies continued to model the impact of the legislation in KEDNY and the Rockaways in KEDLI's service territory per LL 154 specifications."

Q. Does this Panel believe this approach is appropriate?

A. We contend that this approach is generally appropriate. While there may be minor differences between LL 154's requirements and the AEB Act, they are substantively aligned and accounting for both simultaneously would likely duplicate the effect of these laws.

Companies' Forecast Methodology

Q. What methodology did the Companies use to develop their gas delivery volume forecasts?

A. The Companies forecasted their gas load and meter counts based on econometric and statistical forecast models developed for aggregated Residential Heat, Residential Non-Heat, Commercial, Non-firm Demand Response, Multi-Family, and Other rate groups.

Q. Over what historic period were the Companies' gas forecasting models estimated?

A. The Companies' gas forecasting models were estimated using actual monthly sales data beginning as far back as 2007 for some models. All models were estimated on monthly data through February 2023. Provided in response to DPS-1016, the economic explanatory variables used in some of the Companies' models are from the March 2023 release of Moody's economic forecast.

Q. What software did the Companies use to produce their forecasts?

A. The Companies used the R programming language, or GNU-R, which is a widely used open-source

software environment for statistical computing. The Companies also utilized the R-Studio integrated development environment. This integrated development environment provides a number of advanced features for R users, including syntax highlighting and inline visualizations. This allows for the creation of a document that performs and reports the analysis.

Q. Does the Panel have concerns with any of the Companies' methodologies?

A. Yes. We are concerned with the Companies' weather normalization methodology, which defines the heating degree day, or HDD, average used to forecast gas deliveries under normal weather conditions, or the weather normal HDDs, as a rolling 30-year average. This Panel recommends using a 10-year heating degree day average for weather normalization.

Q. Why does the Panel recommend using a 10-year weather normalization instead of the Companies' proposed 30-year weather normalization?

A. Compared to a 30-year average weather normalization, a 10-year average weather

normalization more accurately reflects weather trends and is more likely to accurately predict the weather in the forecasting period. A 30-year average incorporates older data that becomes less relevant when changing climate patterns are taken into consideration, making the 30-year average less representative over time. Given the trend of increasing temperatures, using a 30-year weather average could improperly overstate the anticipated number of heating degree days in the Rate Year forecast period, resulting in a less accurate gas delivery forecast. For instance, based on the weather data provided in the Companies' response to DPS-263, a 30-Year HDD average assumes 4,544 HDD for KEDNY and KEDLI, while a 10-year average assumes 4,438 HDD for KEDNY and KEDLI during the Rate Year.

Q. Is there precedent for using a 10-year heating degree day average in developing a gas delivery forecast?

A. Yes. On page 15 of its Order Adopting Recommended Decision with Modifications issued on June 22, 2009 in Cases 08-E-0887, 08-G-0888,

and 09-M-0004, the Commission adopted the practice of using the most recent 10-year averages for purposes of providing heating and cooling degree day inputs to the sales forecasts. The Commission adopted this recommendation per the Recommended Decision in those cases which found that "compared to actual experience over the 15 years ending 2007, the 30-year average overstated heating degree days (and understated cooling degree days) about two-thirds of the time, while . . . the 10-year average now used by [Energy Information Administration of the U. S. Department of Energy] showed no such bias." The Commission also stated its expectation that these averages would be used for forecasting purposes in future rate filings.

Q. Are there any other justifications for using a 10-year weather average?

A. Yes. In December 2019, the New York Independent System Operator, or the NYISO, issued a Climate Change Impact Study which found that the statewide average temperature has increased by 1.1 degrees on average per decade beginning in

1992. The study specifically anticipates that average temperatures in Central Park in New York City will increase by 0.6 degrees per decade. According to the Companies' response to DPS-263, the Companies calculate their HDDs based on weather data from the Central Park, New York weather station.

Staff's Forecasting Methodology

Q. Did the Panel develop its own gas forecasting models?

A. Yes. We also used econometric models to forecast meter count and use-per-customer for the aggregated residential non-heating, residential heating, commercial, multi-family building, and non-firm demand response rate groups from the data the Companies provided in response to DPS-1016. We then used the allocation factors provided by the Companies in response to DPS-1016 and embedded within their workpapers to disaggregate the Panel's forecasts into each respective KEDNY and KEDLI service classification.

Q. Please summarize which rate classes these rate groups represent.

A. We forecasted the same rate groups the Companies identified on page 17 of the Gas Forecasting Panel direct testimony. For KEDNY, the Residential Non-Heating rate group represents SC 1A and SC-17-1A; the Residential Heating rate group represents SC 1B and SC 17-1B; Commercial represents SC 2-1, SC 2-2, SC 17-2-1, and SC 17-2-2; Multi-Family represents SC 3 and SC 17-3; and Non-Firm Demand Response represents SC 22 and SC 18-22. For KEDLI, the Residential Non-Heating rate group represents SC 1A and SC 5-1A; the Residential Heating rate group represents SC 1B and SC 5-1B; Commercial represents SC 2A, SC 2B, SC 5-2A, and SC 5-2B; Multi-Family represents SC 3 and SC 5-3; and Non-Firm Demand Response represents SC 18 and SC 19.

Q. Are there any rate groups this Panel did not forecast?

A. Yes, the Panel did not forecast the "Other" rate groups for KEDNY and KEDLI.

Q. Why did this Panel not forecast the "Other" rate groups for KEDNY and KEDLI?

A. For both Companies, the "Other" rate group is relatively small and did not have a strong fit

to our statistically-based models. The Companies, who monitor the sales of these customers and have relationships with them, are better suited to forecast the future sales of this rate group.

Q. Please describe the basic functioning of an econometric model.

A. An econometric model relates a dependent or response variable, in this case gas sales per customer or the number of customers, to a set of explanatory or independent variables. These explanatory variables include weather, economic, demographic, and other seasonal and cyclic variables. Weather variables are represented by heating degree days. Seasonal variables may adjust for calendar month or other billing-cycle-related variations. The economic and demographic explanatory variables are the structural components of the econometric model.

Q. Please describe the non-structural, time-series components of the forecast models.

A. The non-structural component is represented by a procedure that relates the present time series values and model estimation errors to the

historical time series values and model estimation errors. This is performed through the inclusion of moving average, or MA, and autoregressive, or AR, variables to the model specification. We include these non-structural components to capture the remaining variations in sales or meter counts, including time trends and seasonality in gas sales or number of customers, which are not explained by the structural explanatory variables. The actual data for per customer sales or meter counts are used as the dependent variables, and the independent variables include actual values for HDD, economic, and demographic variables. We developed econometric models by rate group for meter counts and use-per-customer. Our models are estimated using updated historical data through February 2023. For those service classes for which we did not use a structural econometric model, we forecasted sales using a purely time-series model (i.e., non-structural model that relates the present time series values and model estimation errors to the historical time series values and model

estimation errors).

Q. Can the Panel identify the forecasting assumptions for the explanatory variables in its gas models?

A. For weather, we assume the 10-year average ending December 31, 2021 as normal weather which the Companies provided in response to DPS-263. For economic variables, we use the March 2023 Moody's forecast provided by the Companies in response to DPS-1016.

Q. Please discuss the Panel's aggregated Residential Non-Heat, Residential Heat, Commercial, Non-Firm Demand Response, and Multi-family forecasting models.

A. Our Residential Non-Heat, Residential Heat, Commercial, Non-firm Demand Response, and Multi-family forecasts are each based on two models, a meter count forecast and a use-per-customer forecast for KEDNY and KEDLI. The forecasted outputs from these models are combined to produce monthly billed therm sales as their final result. The explanatory variables in the models include the heating degree days, economic, and demographic variables. For each

of the models, we estimate using a minimum of five years monthly historical data through February 2023, which was provided by the Company in response to DPS-1016.

Q. Starting with KEDNY, please compare this Panel's forecasts and models with the Companies' by rate group.

A. All of KEDNY's forecasts are constructed from two sub-models, combining a rate group meter count forecast, which is then converted to customer accounts, and a use-per-customer model. The Panel's models follow a similar structure.

Q. What is the implication of constructing the forecasts in this way?

A. The implication of the construction of these forecasts in this way implies that the usage-per-customer is the same for all customers. Since this is not a reasonable assumption, as usage among customers varies, we interpret the meter count forecast as the average number of customers, that is meters before conversion, for the group being forecasted. In this formulation, the inclusion of meter counts in any use-per-customer model can be interpreted as

the rate at which the average usage changes as the number of customers change.

Staff's Forecasts for KEDNY

Residential Non-Heat and Heat Forecasts

Q. Please describe both the Panel's and the Companies' residential customer and sales forecasts for KEDNY.

A. Starting with residential non-heat customers and sales, KEDNY forecasts an average of 552,020 customers and total volumes of 54,216,365 therms for the Rate Year. We forecast an average of 561,870 customers for the Rate Year, with total sales volumes of 55,995,124 therms for the Rate Year.

Q. Please describe KEDNY's residential non-heat meter and use-per-customer forecast models.

A. KEDNY's residential non-heat meter count model is a regression model with Autoregressive Moving Average, or ARMA, errors driven by a trend, a binary variable representing the period of August 2020 through October 2021, representing the height of the COVID-19 pandemic, and an autoregressive term that is dependent on the previous month's meter counts, or an AR(1) term.

KEDNY's use-per-customer model is similar, also a linear regression model with an AR(1) and an SMA(1) term, driven by the Companies' definition of billing degree days, or BDD.

Q. What are billing degree days?

A. BDDs are the sum of the daily HDD values from mid-month to mid-month to reflect the lag inherent in billing volumes.

Q. What concerns, if any, does the Panel have with KEDNY's residential non-heat model formulation?

A. We do not have any specific concerns with the Companies' specifications but argue they can be improved with some modifications. We defined a forecasting model very similar to KEDNY's for the residential non-heat meter count forecast. Like KEDNY, our model is a linear regression model with an AR(1) term; however, we used a log-level formulation with the addition of the three-month lagged service territory gross domestic product. Put simply, we added a gross domestic product variable into our econometric model.

Q. Why does the Panel contend its model is more reasonable than the KEDNY's?

A. First, by including service territory gross domestic product, the number of meters now tracks the local economy's growth. With each one percent increase in service territory gross domestic product, forecasted residential non-heat meter counts will increase by an average of 0.04 percent. As such we contend that our model will yield a more reasonable forecast that reflects economic conditions by linking changes in meter counts to economic conditions.

Q. Please discuss KEDNY's residential non-heat use-per-customer model.

A. KEDNY's use-per-customer model is similar to its meter count model, and is also a linear regression model with an AR(1) and a Seasonal Moving Average(1) term. Their model is also driven by the Companies' definition of billing degree days. KEDNY's and our residential non-heat use-per-customer models are again very similar, with the primary difference being our use of a log-level formulation, and the use of the 10-year normal billing degree days instead of the 30-year normal billing degree days used by KEDNY. As described previously, we contend

the 10-year normal billing degree day measure is more accurate than the 30-year normal given the changing climate.

Q. What is this Panel's recommendation regarding the residential non-heat sales forecast?

A. We recommend of the Commission adopt our residential non-heat sales forecast for the reasons explained previously.

Q. Please describe the forecasts for the residential heat rate group.

A. For residential heat customers, KEDNY forecasts an average of 661,400 customers and total volumes of 756,100,775 therms. We forecast an average of 670,361 customers and total volumes of 764,984,761 therms.

Q. Please describe KEDNY's residential heat meter and use-per-customer models.

A. The Company's residential heat meter model is defined as a regression model with AR(1), AR(2), MA(1), MA(2), and SAR(1) terms. An MA term is a moving average, an AR is an auto-regressive term and an SAR term is a seasonal auto-regressive term. The model is driven by the housing stock economic variable. KEDNY's residential heat

use-per-customer model is also a linear regression model but does not contain any ARIMA component. It is primarily driven by billing degree days using the 30-year weather normalization methodology.

Q. What concerns, if any, does the Panel have regarding the Company's formulation of its residential heat models?

A. Our primary concern with the Company's models is its use of the 30-year weather normalization methodology. Our residential heat meter count model is similar to the KENDY's meter count forecast model but modified with a log-level structure and a three-month lag in housing stock. Our model similarly uses BDD, but again uses the 10-year weather normalization methodology instead of the 30-year weather normalization methodology. Our model also adds the two-month lag of residential natural gas prices.

Q. Why does the Panel contend its model is more reasonable than KEDNY's?

A. We contend that economic variables theoretically drive sales, and therefore, should be included

in the forecasting models when reasonable. The inclusion of the price of natural gas and our use of the 10-year weather normalization methodology are indicative of this principle and leads us to conclude that our models are more reasonable than KEDNY's.

Q. What is this Panel's recommendation regarding the residential heat sales forecast?

A. This Panel recommends that the Commission adopt this Panel's meter and volume forecasts for the reasons explained above.

Commercial Forecast

Q. Please describe the forecasts for the commercial rate group.

A. KEDNY forecasts an average of 52,597 customers with total volumes of 293,647,325 therms. We forecast an average of 52,837 customers for the Rate Year with total volume sales of 303,109,942 therms.

Q. Please describe KEDNY's commercial meter and use-per-customer models.

A. The Company's commercial meter count forecast is a linear regression model with an AR(1) and AR(2) terms, an MA(1) term, and an SAR(1). The

model is driven by a one-month lag in housing stock, and includes a monthly varying constant term to account for month-to-month variation. The Company's use-per-customer model is another linear model that does not include any AR or MA terms and is driven by BDD, KEDNY's service territory's gross domestic product, and the change in commercial use-per-customer lagged by two months.

Q. What issues, if any, does the Panel have with KEDNY's model formulations?

A. Our primary concern with these models is the usage of the 30-year normalization methodology in the calculation of the billing degree days variable in the Company's use-per-customer model. We also made alternative modeling choices for meter counts and use-per-customer.

Q. Please describe the Panel's commercial meter and use-per-customer models.

A. Similar to KEDNY's meter count models, our model of commercial meter counts features a varying-by-month constant term to account for month-to-month fluctuations. Instead of using housing stock as a driver of commercial meters, our

model uses a simple trend to capture the average growth rate and the current population in the service territory. The expectation is that a larger population will require more commercial entities, resulting in more commercial meters.

Q. Please compare and contrast KEDNY's and the Panel's use-per-customer models.

A. Again, like KEDNY, we utilized BDD to drive our model but with 10-year weather normalization methodology instead of the Companies' 30-year weather normalization methodology. We also replaced KEDNY's service territory's gross domestic product variable with a time trend, averaging out commercial growth and then accounting for weather variations.

Q. What are is the Panel's recommendation regarding KEDNY's commercial meter and use-per-customer forecasts?

A. We recommend that the Commission adopt our models and forecasts as we contend that population is a more reasonable driver than housing stock for commercial properties because during economic downturns or other challenging situations, renting or sharing domiciles becomes

more common, while commercial demand will continue to grow as the population grows. Further, the usage of the 10-year weather normalization methodology ensures a more accurate accounting for changes in the climate.

Multi-Family Forecast

Q. Please describe the forecasts for the multi-family home rate group.

A. KEDNY forecasts 19,939 customers with total volume sales of 387,300,392 therms in the Rate Year. We forecast 19,923 customers with total volume sales of 398,294,646 therms.

Q. Please describe KEDNY's meter count and use-per-customer models.

A. KEDNY's meter count forecast is a linear regression model with an AR(1) term. Their model is driven by housing stock. KEDNY's model's use-per-customer sub-model is also a linear regression but does not include any AR or MA terms. KEDNY also uses BDD, income-per-capita, the current change in income-per-capita from the previous month, the change in BDD from the previous month, and the three-month lagged change in multi-family home use-per-customer to

drive their model.

Q. How do these variables compare to the Panel's forecast models?

A. Like KEDNY, our meter count forecast is a linear regression model with AR and MA terms, specifically an AR(1), AR(2), and a MA(12) terms; however, instead of using housing stock as an economic driver, we instead use a one-month lag of population.

Q. Please discuss this Panel's multi-family housing use-per-customer model.

A. Our multi-family home use-per-customer model is also a linear regression model with AR(1), AR(12), MA(1), and MA(12) terms, but is overall simpler than KEDNY's. To model this group's usage, we utilized BDD based off the 10-year weather normalization methodology multiplied by a trend variable. Use of the 10-year average of BDDs reflects our belief that the primary driver of usage-per-customer in this class is weather, but as technology improves and access to efficiency measures and electrification of large buildings continues to increase, customer's willingness and ability to change their usage

increases.

Q. What is this Panel's recommendation regarding KEDNY's multi-family meter and use-per-customer forecasts?

A. For the reasons explained above, we recommend that the Commission adopt our models and forecasts.

Non-firm Demand Response Forecast

Q. Please discuss KEDNY's forecasts for the non-firm demand-response rate group.

A KEDNY forecasts 1,771 customers with total volume sales of 101,865,678 therms. We forecast 1,894 customers with total volume sales of 105,612,313 therms.

Q. Please describe KEDNY's models for non-firm demand response meter counts and use-per-customer forecasts.

A. The model used in KEDNY's non-firm demand response meter count forecast is a stark departure from their other models. It is a naïve drift model.

Q. Can the Panel describe what a naïve drift model is?

A. A naïve drift model takes two values, the last

observed value and the calculated drift, defined as the average change between consecutive data points over the entire data period. For each period, the last observed value is adjusted by the average change per period, multiplied by the number of periods between the observation and the current period.

Q. Please describe KEDNY's use-per-customer model for the non-firm demand-response rate group.

A. KEDNY's use-per-customer model is more comparable to its other models, as it is a linear regression model with an SAR(1) term driven primarily by BDD.

Q. Does the Panel have any specific concerns about the Company's forecast formulation?

A. Yes. We are concerned that the Company used a naïve drift to forecast the non-firm demand-response meter count. We argue that a more rigorous model, like those used by the Panel and the Company elsewhere in its testimony, including its non-firm demand-response use-per-customer model, would produce a more reasonable forecast.

Q. How did the Panel approach modeling forecast

meter counts for non-firm demand response customers?

A. Our non-firm demand response meter count model is a linear regression model an AR(1) and MA(1) and is primarily driven by a trend variable.

Q. How does the Panel model the non-firm demand response group's use-per-customer forecast?

A. Our non-firm demand response use-per-customer model is similar to KEDNY's, as it is primarily driven by BDD. The crucial difference, as is the case with all the models discussed previously, is the use of a 10-year weather normalization methodology instead of a 30-year weather normalization methodology.

Q. What is the Panel's recommendation regarding KEDNY's non-firm demand-response meter count and use-per-customer forecasts?

A. We recommend that the Commission adopt our models and forecast for this group.

Other Forecast

Q. Please continue with the Other rate group.

A. Due to its relatively small size, we did not perform an alternative econometric forecast for the Other rate group. For the Rate Year, KEDNY

forecasts an average of 177 meters with total volumes of 401,271,099 therms.

Staff's Forecasts for KEDLI

Q. Please compare and contrast this Panel's sales forecasts with KEDLI's.

A. As with the KEDNY forecasts, all KEDLI's forecasts are constructed from two sub-models, combining a rate group meter count forecast and a use-per-customer model. Unlike KEDNY, KEDLI does not have a meter-to-customer account conversion rate, so we assume each meter corresponds to one customer account.

Q. What is the implication of KEDLI constructing their forecasts in this way?

A. The implication of the construction of these forecasts in this way implies that the usage-per-customer is the same for all customers. Since this is not a reasonable assumption for the reason explained above, we interpret the meter count forecast as the average number of customers for the customer group. In this formulation, the inclusion of meter counts in any use-per-customer model can be interpreted as the rate at which the average usage changes as

the number of customers change.

Residential Non-Heat and Heat Forecasts

Q. Please describe KEDLI's Residential Non-Heat forecasts.

A. Starting with residential non-heat customers and sales, KEDLI forecasts an average of 73,339 customers and total volumes of 14,770,301 therms for the Rate Year. We forecast an average of 72,702 customers for the Rate Year, with total sales volumes of 14,867,496 therms for the Rate Year.

Q. Please describe KEDLI's residential non-heat forecast and its statistical and theoretical underpinnings.

A. KEDLI's residential non-heat meter count model is a linear regression model with an AR(1), MA(1), and SMA(1) terms. It is primarily driven by a trend variable and does not include any economic or demographic related drivers. KEDLI's residential non-heat use-per-customer forecast is also a linear regression model and is primarily driven by BDD.

Q. What issues, if any, does the Panel have with KEDLI's residential non-heat forecast?

A. Our primary concern for KEDLI's residential non-heat forecast is the use of 30-year normalization methodology in the calculation of BDD.

Q. How does the Panel model the KEDLI residential non-heat rate group forecasts?

A. Our approach is largely similar to KEDLI, in that we use a linear regression with an AR(1) and MA(1) terms. This meter count model is primarily driven by a trend. Our use-per-customer models are linear regressions with an AR(1) and SAR(12) terms, and is primarily driven by BDD but calculated using a 10-year weather normalization methodology.

Q. Why does the Panel contend its forecast is more reasonable than KEDLI's?

A. As described previously, we contend that the 10-year weather normalization methodology better reflects the change in climate that is occurring. Therefore, this Panel recommends the Commission adopt our residential non-heat meter count and residential non-heat use-per-customer forecasts.

Q. Please describe the forecasts for residential

heat customers for KEDLI.

A. For residential heat customers, KEDLI forecasts an average of 497,890 customers and total volumes of 605,451,660 therms. We forecast an average of 498,999 customers and total volumes of 597,093,859 therms.

Q. Please describe KEDLI's forecast model for Residential Heat.

A. KEDLI's residential heat meter count model is very similar to their non-heat meter count model, consisting of a basic linear regression model with AR(1), MA(1), and SMA(1) terms. The primary driver of this model is a trend variable. KEDLI's use-per-customer model is also largely similar to its residential non-heat use-per-customer model, being a linear regression without an AR or MA terms.

Q. What issues, if any, does the Panel have with KEDLI's forecast formulation?

A. As with KEDLI's residential non-heat forecast, Our primary concern is the use of BDDs calculated with 30-year weather normalization methodology.

Q. How does the Panel model the residential heat

rate group?

A. Like KEDLI, we modeled residential heat meter counts as a linear regression model with an AR(1) and MA(1), however, we also included housing stock as an economic explanatory variable. For the use-per-customer model, we also used BDD as the primary driver, but used a 10-year weather normalization methodology, along with an AR(12) term and a MA(12) term.

Q. What additional insights from the Panel's formulation of these variables can be made?

A. The inclusion of housing stock helps the model of meter counts follow the economic climate and produces a statistically significant impact to the model. Additionally, there is evidence that our meter count model fits the data better than the Company's as it achieves an adjusted r-square of 0.9893 as opposed to KEDLI's model, which has an adjusted R-square of 0.9761. Therefore, the Commission should adopt this Panel's forecast.

Commercial Forecast

Q. Please describe KEDLI's forecasts for the commercial rate group.

A. KEDLI forecasts an average of 60,933 customers with total volumes of 417,611,643 therms. We forecast an average of 61,061 customers for the rate year with total volume sales of 416,839,978 therms.

Q. Please describe KEDLI's commercial meter count and use-per-customer forecast and its characteristics.

A. KEDLI's commercial meter count forecast is built off of a linear regression model and an AR(1), AR(3), and MA(1) terms. It has a monthly varying constant term to account for monthly fluctuations in meter counts. It does not contain any other explanatory variables. KEDLI's commercial use-per-customer model is very similar to its residential use-per-customer models and is a linear regression model driven by BDD and a trend variable without any ARIMA terms.

Q. What issues, if any, does the Panel have with KEDLI's commercial forecasts?

A. As is the case with both Companies' models, Our concern is the usage of BDD calculated with a 30-year weather normalization methodology.

Q. How did the Panel approach modeling the commercial rate group?

A. Our models are similar to KEDLI's with modification. For our meter count forecast we used a linear regression model with AR(1), AR(3), and MA(1) terms along with a monthly-varying constant term. We also added a trend variable to capture the persistent growth found in the data. For our use-per-customer forecast, we also included a persistent trend, as well as utilizing BDD calculated with 10-yr normal and an AR(11) and MA(1) terms.

Q. Why does the Panel contend its forecasts are more reasonable than KEDLI's?

A. We contend that the usage of our BDD 10-year weather normalization methodology is a more accurate way to include weather affects in the model. Therefore, this panel recommends the Commission adopt our forecast.

Multi-Family Forecast

Q. Please describe the forecasts for the multi-family home rate group.

A. KEDLI forecasts an average of 1,775 customers with total volume sales of 53,107,620 therms in

the rate year. We forecast an average of 1,765 customers with total volume sales of 52,627,859 therms.

Q. Please describe KEDLI's multi-family home rate group forecast.

A. KEDLI's multi-family home meter-count forecast is built upon a linear regression model with an AR(1) and SAR(1) terms. It is driven by a general trend variable. KEDLI's use-per-customer model is a linear regression model without AR or MA terms and is driven primarily by BDD.

Q. How does the Panel approach modeling multi-family home rate group?

A. Our model of multi-family home meter counts is a linear regression model with an AR(1) and an SAR(12) term. It also includes housing stock as a driver of meter counts. For our use-per-customer model, it is similar to KEDLI's but utilizes BDD, a 10 year weather normalization, and includes an MA(11) and MA(12) term.

Q. Why does the Panel contend its forecast is reasonable than KEDLI's?

A. In addition to utilizing the appropriate BDD

formulation, we contend that our forecast is more reasonable because it includes an economic explanatory variable in its meter count forecast, and the use-per-customer model produces a better fit, with an adjusted R-square of 0.9793 vs. the Company's model which achieves an adjusted R-square of 0.9581.

Q. What is this Panel's recommendation regarding multi-family home forecast?

A. This panel recommends the Commission adopt our forecast.

Non-Firm Demand-Response Forecast

Q. Please describe the forecasts for the non-firm demand-response rate group.

A KEDLI forecasts an average of 170 customers with total volume sales of 12,793,082 therms. We forecast an average of 171 customers with total volume sales of 13,274,616 therms.

Q. Please describe KEDLI's non-firm demand response rate group forecast.

A. The Company's non-firm demand response meter count forecast was implemented with a naïve random walk, with no persistent drift. Colloquially, this can be understood as if KEDLI

were to flip a coin to determine whether meter counts will be higher or lower in the next month, as compared to the current month. The Company's use-per-customer is a linear regression model driven by BDD and includes an AR(1) and MA(1) terms.

Q. What issues, if any, does the Panel have with KEDLI's non-firm demand response rate group forecast?

A. We have a few concerns. First, we contend that the meter count forecast should be modeled more deterministically. Our meter count forecast achieves an adjusted R-square of 0.9989. We also continue to disagree with the Company's use of BDD calculated using a 30-year term to reflect normal weather. Therefore, we recommend that the Commission adopt our meter count forecast.

Q. How should the Commission address these concerns?

A. The Commission can address these concerns by adopting our non-firm demand response rate group forecasts as they are more reasonable than KEDLI's and rely upon explanatory macroeconomic

variables rather than chance.

Other Group Forecast

Q. Please describe the forecasts for the Other rate group.

A. Due to its relatively small size, we are not able to forecast this rate group econometrically with a reasonable level of precision. Further, the small sample size also suggests that other factors, which could not be readily incorporated into an econometric model, are major drivers of sales in this rate group. Thus, we did not perform an alternative econometric model for the Other rate group. For the rate year, the Company forecasts an average of 13 meters, with total volumes of 197,109,305 therms.

Summary Statics of Staff's Forecasting Models

Q. Please discuss the summary statistics of the Panel's estimated forecasting models.

A. As shown in Exhibit__(SFIP-4), all the models have high adjusted R-squared with values. The R-squared statistic, ranging from zero to one, measures the degree of goodness-to-fit of the regression model, with zero being the least fit and one being the best fit. The adjusted R

squared of the 20 models ranges from 0.87 to 0.99, suggesting that approximately 87 to 99 percent of variations in deliveries and customer counts are explained by the included explanatory variables. The Durbin-Watson statistics are between 1.81 and 2.18 indicating that the models do not exhibit strong autocorrection. These summary statistics show that our forecasting models perform well and have met the standard criteria used for evaluating the reasonableness of estimated econometric models.

Out of Model Adjustments

Q. Did the Panel make post-model adjustments to its gas volume model forecasts?

A. We have used the same post-model adjustments that have been used by the Companies. The Companies summarize these adjustments in Exhibit___(GLF-15ACU), Exhibit___(GLF-15BCU), Exhibit___(GLF-16ACU), and Exhibit___(GLF-16BCU).

Q. Please describe the Companies' Gross MMBtu targets for the Rate Year.

A. According to Appendix A – Table A4 of the Commission Order Authorizing Utility Energy

Efficiency and Building Electrification Portfolios Through 2025 issued and effective January 16, 2020 in Case 18-M-0084, the KEDLI Gross MMBtu targets for years 2024 and 2025 are 953,821 and 1,129,821, respectively. The KEDNY Gross MMBtu targets for years 2024 and 2025 are 1,082,740 and 1,347,740, respectively. Since the Rate Year falls within 75 percent of year 2024 and 25 percent of 2025, the KEDLI and KEDNY targets would be 997,821 MMBtu and 1,148,990 MMBtu, respectively, for the Rate Year.

Q. How does this compare with the Companies' post model adjustments provided in Exhibit___(GLF-16ACU) and Exhibit___(GLF-16BCU) that they are projecting based on their expected achievement of the New Efficiency New York goals?

A. The calculated Gross MMBtu targets for the Rate Year are the KEDLI and KEDNY targets would be 997,821 MMBtu and 1,148,990 MMBtu, or 9,978,210 therms and 11,489,900 therms, respectively. According to Exhibit___(GLF-16ACU) and Exhibit___(GLF-16BCU), the total anticipated savings from energy efficiency (above historical trend) will be 9,136,710 therms and 11,587,244

therms for KEDLI and KEDNY, respectively.

Q. Are these targets aligned with the targets set by the Commission?

A. Since these targets are based on a calendar year, the Companies projections appear to be aligned with the Commission ordered targets.

Inflation

Q. Please describe the Companies' non-labor inflation calculation.

A. As detailed in the Companies' Exhibit___(RRP-8), pages 125 through 126 of KEDNY's of RRP's direct testimony, and pages 126 through 127 of KEDLI's RRP's direct testimony, the Companies used the Chained Price Index Gross Domestic Product Implicit Price Deflator, or GDPIPD, to represent the level of non-labor input prices and percent changes in this index as the measure of inflation. The Companies obtained forecasted values of the GDPIPD on March 14, 2023 from Moody's Analytics. The Companies relied upon historical values of the GDPIPD from quarterly data released by the United States Department of Commerce, Bureau of Economic Analysis, or BEA, and calculated inflation from the HTY to the

Rate Year by dividing the average of the forecasted GDPIPD in the Rate Year by the average of the GDPIPD in the HTY. The Companies used the same methodology to calculate inflation between the Rate Year and the subsequent FYs.

Q. Please provide the non-labor inflation rates the Companies calculated for the Rate Year and each of the subsequent FYs.

A. The Companies' calculation of inflation is 6.54 percent from the HTY to the RY, 1.93 percent for FY26, 1.81 percent for FY27, and 1.77 percent for FY28.

Q. Did the Companies update their inflation calculations in their Corrections and Updates filing on June 30, 2023?

A. Yes. Exhibit___(RRP-8CU), shows the Companies' updated calculations of inflation are 6.59 percent from the HTY to the RY, 1.94 percent for FY26, 1.83 percent for FY27, and 1.78 percent for FY28.

Q. Did the Companies use the same methodology as the Panel previously discussed to calculate the inflation rates in their Corrections and Updates filing?

A. Yes, however, the Companies relied upon updated forecasted values of the GDPIPD in years 2023 through 2028 that they obtained from Moody's Analytics on May 9, 2023.

Q. Does the Panel agree with the Companies' calculation of its non-labor inflation rates?

A. No, we do not agree with the Companies' use of Moody's Analytics as the source of the forecasted values of the GDPIPD. Instead, we recommend that inflation be calculated using the most recent consensus forecasts of the Gross Domestic Product Chained Price Index, or GDP-PI, included in the August 11, 2023 Issue of the Blue Chip Economic Indicators.

Q. Please explain why the Panel recommend using the GDP-PI from the Blue Chip Economic Indicators as opposed to the GDPIPD values from Moody's Analytics as the source of the forecasted values.

A. As explained in the Notice Requesting Comments on a Proposed Statement of Policy in Regard to Escalation Rates in Rate Cases issued in Case 27758 on May 22, 1980, the Commission wished to adopt a consensus measure of inflation "because

it believe[d] that an inordinate amount of time and effort [was] being spent by parties in rate cases arguing over what the proper measure of inflation should be in making expense projections. The Commission [did] not believe that the contribution made to the decisional record by this debate justifie[d] the cost." Accordingly, as subsequently established by the Commission in its Notice of Proposed Change in the Index Used to Measure Inflation for Use in Rate Making Proceedings, issued on April 1, in Case 92-M-0184, the Commission should continue to follow its long-standing policy to use a consensus forecast, more recently the Blue Chip Consensus forecast, for calculating inflation in rate cases and other Commission proceedings.

Q. Please summarize the Panel's recommendation to the Commission regarding the calculation of inflation.

A. As stated before, we recommend that inflation be calculated based on the latest available quarterly inflation forecasts in the Blue Chip Economic Indicators issue published August 11, 2023, and the latest available long-term annual

forecasts in the Blue Chip Economic Indicators issue published March 10, 2023. Based on a weighted average calculation using these latest available forecasts, we recommend that inflation be projected as an increase of 6.82 percent from HTY to RY 1, 2.1 percent for FY26, 2.13 percent for FY27, and 2.175 percent for FY28. These calculations are presented in Exhibit___(SFIP-6).

Q. Was this latest inflation rate reflected in Staff's revenue requirements?

A. No, we are not updating the inflation rate at this time, given the slight differential between the Companies' and our inflation factor. However, we recommend that the estimate of inflation will be updated prior to the conclusion of this proceeding using the Blue-Chip Consensus forecast to determine the Rate Year inflation factor. Thus, we utilized the 6.59 percent inflation forecast provided in the Companies' corrections and updates filing.

Q. Does this conclude the Panels' testimony at this time?

A. Yes.